Piano playing in quiet and in pink noise: a single-subject fNIRS pilot (LYDIA)

Adam Emile Aske

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1 Methods

1.1 Participant, task and sessions

A student pianist (sub-01, graduate level) played Gymnopédie no. 1 (Erik Satie) on a piano in a music recording studio hall. In the NOISE condition a loudspeaker placed behind and to the left of the player emitted pink noise. In QUIET the room was silent. Sessions were run in the order `quiet_1` (ses-08), `noise_1` (ses-09), `quiet_2` (ses-10), `noise_2` (ses-11), all on 2026-09-07. Each session was one run of 6 play blocks (planned 40 ± 8 s, jittered) separated by rest (30 ± 5 s), preceded by a 60 s baseline. A 1 kHz beep cued each play (1 beep) and rest (2 beeps) onset. Markers were sent over the Lab Streaming Layer (LSL) with codes 20 = play(QUIET), 21 = play(NOISE) and 30 = rest. Executed play blocks lasted 32–46 s (mean 38.4 s, `data/blocks.csv`); the design of the four runs is shown in Fig. 1.

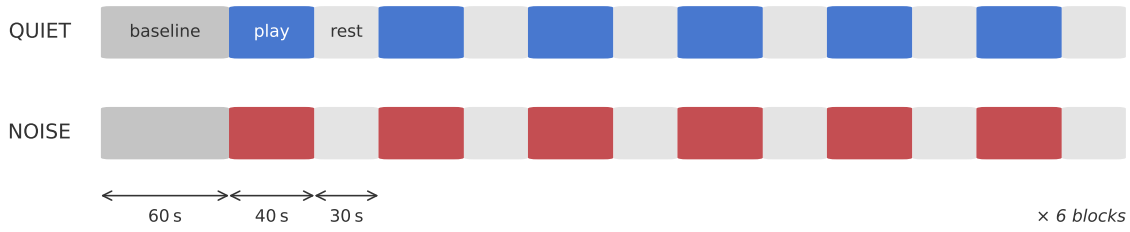


Figure 1: Block design (schematic). Each run began with a 60 s baseline and alternated 6 play blocks with rest; the QUIET and NOISE runs share the same structure. Play and rest durations were jittered around the nominal 40 and 30 s shown here (executed play blocks 32–46 s), and the GLM modelled every block as a 40 s boxcar.

1.2 Montage and acquisition

Functional near-infrared spectroscopy (fNIRS) data were acquired with a NIRSport2 (Aurora 2025.2) at 760/850 nm and 12.2 Hz. The optode layout (Fig. 2) uses 16 sources and 16 detectors on EEG 10-10 sites, giving 47 channels of 30–42 mm separation (mean 35.4 mm). Coverage: frontal pole and ventrolateral prefrontal cortex (VLPFC; Fpz, AF7, AF8 → Fp1, Fp2, AFz), dorsolateral prefrontal cortex (DLPFC) (AF3, AF4, Fz, F3, F4 → F5, F1, F2, F6), premotor and supplementary motor area (SMA; FC5, FC1, FC2, FC6 → FC3, FCz, FC4), sensorimotor (C3, C4 → C5, C6) and temporal (T7, T8 → FT7, TP7, FT8, TP8).

1.3 fNIRS preprocessing

The raw data files were copied and the play markers set to a 40 s duration (executed mean 38.4 s). Preprocessing was performed in NIRWizard (0.1.7) using its Cedalion (v26.5.2) wrapped preprocess-

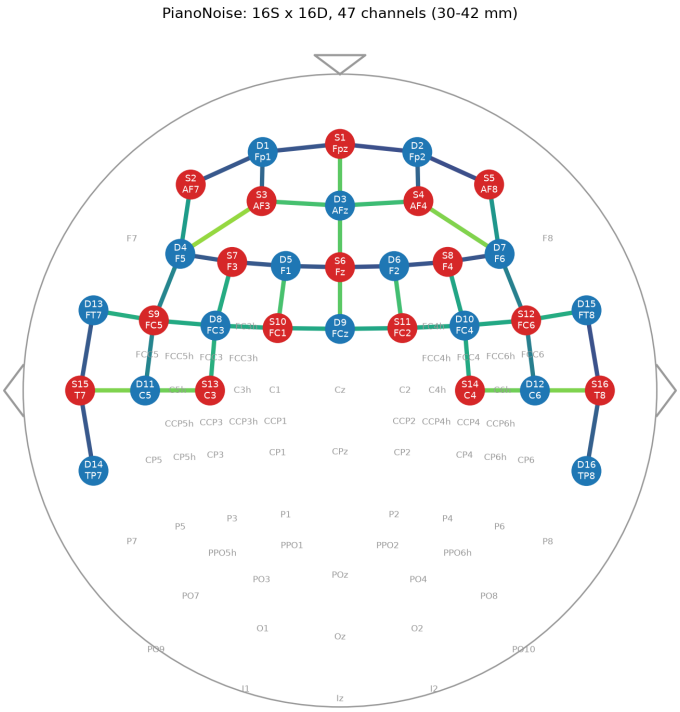


Figure 2: The optode layout, top view (nose up). Red: sources S1–S16, blue: detectors D1–D16, lines: the 47 channels (colour = source–detector distance, 30–42 mm).

ing module. The four recordings (quiet_1, noise_1, quiet_2, noise_2) were treated as four runs of one session (concatenated). Raw intensities at 760 and 850 nm were converted to optical density (OD) relative to the temporal mean of each channel. Motion artefacts were corrected in three steps: principal component analysis (PCA) artefact removal on segments detected by amplitude/standard-deviation thresholds (`tMotion`=0.5 s, `tMask`=1 s, STDEV 50, AMP 5; 97 % of variance, 15 components, 9 segments of 2–5 s), followed by temporal derivative distribution repair (TDDR) and wavelet-based correction (db2, 4 levels, interquartile range 1.5). The corrected OD signals were band-pass filtered between 0.01 and 0.5 Hz (4th-order zero-phase Butterworth) to remove slow drift and cardiac components. Concentration changes of oxy- and deoxyhaemoglobin (HbO, HbR; μM) were computed with the modified Beer–Lambert law (Scholkmann & Wolf 2013; age 25 assumed; differential pathlength factor 6.15 / 5.09 at 760 / 850 nm). The data were resampled from 12.2 to 10 Hz (anti-aliased) to resolve a NIRWizard–Cedalion mismatch, giving 21 429 samples (2 143 s). Channel quality was assessed with windowed scalp-coupling index (SCI) and peak spectral power (PSP), but no channel was excluded from the analysis reported here (Section 2.4).

1.4 General linear model

Subject-level general linear model (GLM) with one boxcar regressor per condition (QUIET, NOISE; event duration 40 s) convolved with the canonical SPM haemodynamic response function (HRF; kernel 32 s, peak delay 6 s, undershoot delay 16 s, ratio 6), per-run discrete cosine transform (DCT) drift terms (cutoff 128 s, 31 columns), three run-intercept columns and a global intercept. The solver was autoregressive iteratively reweighted least squares (AR-IRLS; Barker, Aarabi & Huppert 2013): Yule–Walker autoregressive prewhitening with Bayesian information criterion order selection up to 40 samples (4 s; selected orders 8–33), pooled over runs and restarted at every run seam, with Tukey-bisquare robust weights. Inference is per channel on the canonical-HRF β with df $= n - p = 21\,392$ and Benjamini–Hochberg false discovery rate (FDR) control within each condition $\times$ chromophore family. No QUIET-vs-NOISE contrast was fitted, so the two conditions are compared descriptively.

Cortical renders were exported from NIRWizard (GLM t, HbO, cortex opaque, probe hidden,

800×800) in two versions: unthresholded, and with a descriptive display threshold $|t| > 10$. With df $\approx 21\,000$ the t values are large everywhere. The $|t| > 10$ cut only picks out the strongest channels and is not a significance level.

2 Results

2.1 Play-related activation in both conditions

Playing increased HbO almost everywhere the montage looks: 45/47 channels in QUIET and 44/47 in NOISE survive $q < 0.05$, with positive β in all but a few channels (up to 0.41 μM). HbR is far weaker (27 and 30 channels at $q < 0.05$, mixed sign, $|\beta| \leq 0.1\,\mu$M) and the canonical HbO-up/HbR-down pattern is present in only part of the montage; laterality is therefore reported for HbO. Figure 3 shows the unthresholded HbO t-maps. In both conditions the strongest channels are the frontal-pole row (Fpz–Fp1/Fp2/AFz, t = 11–13), left premotor/motor (FC5–C5, t = 13.5 and 11.5) and left temporal (T7–C5, $t = 10.7$). Both maps look bilateral at this level; the difference between them is one of degree in the right lateral channels.

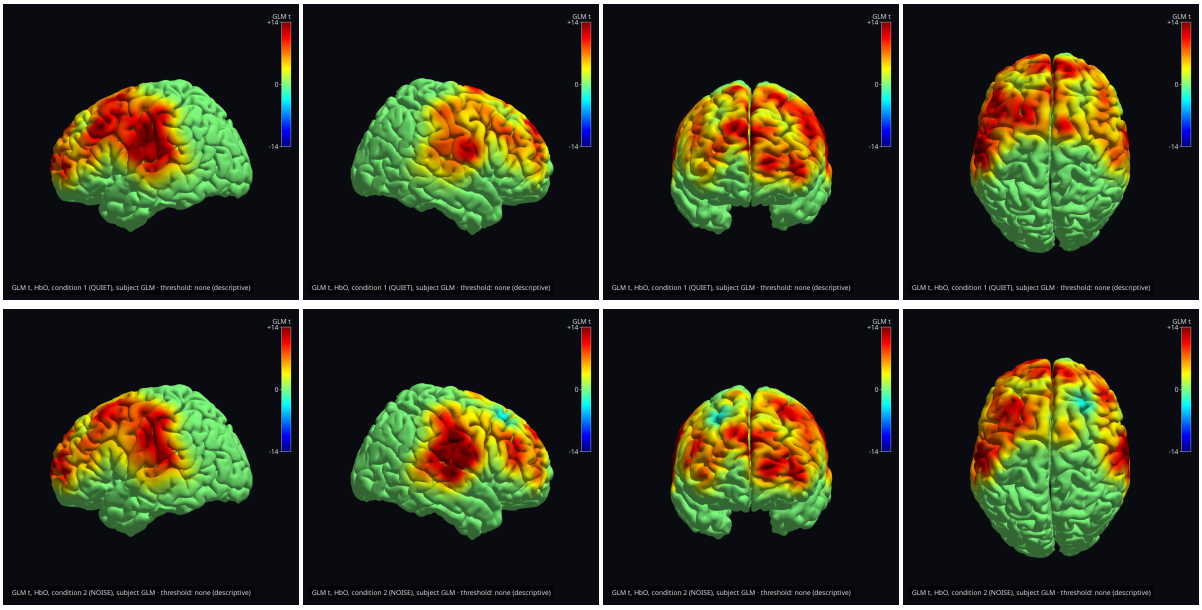


Figure 3: Unthresholded GLM t (HbO) projected on the cortex. Top row: QUIET, bottom row: NOISE. Columns: left lateral, right lateral, anterior, superior view. Colour scale ±14.

2.2 Thresholded maps: a right-hemisphere cluster appears only in noise

At $|t| > 10$ (Fig. 4) exactly 13 HbO channels survive in each condition, but not the same 13 (Table 1). In QUIET, 7 are left-hemisphere, 2 midline and 4 right, and the right ones are all prefrontal (Fpz–Fp2, AF4–AFz, FC2–FCz) plus FC6–FT8. In NOISE, 6 are left, 1 midline and 6 right, and three right channels enter that are far below threshold in QUIET: T8–FT8 (t 5.0 → 12.6, β 0.09 → 0.23 μM), FC6–C6 (5.9 → 12.5, 0.15 → 0.32 μM) and T8–C6 (7.2 → 11.0, 0.13 → 0.20 μM). Together with FC6–FT8 (above threshold in both) they form a contiguous right temporal–ventral premotor strip, visible as the large red patch in the right lateral NOISE view and as the right-lateral extension in the superior view. The neighbouring right channels C4–C6 and AF8–F6 also rise (7.5 → 8.8 and 5.1 → 8.8) without reaching the cut. On the left, the mirror channels do not change or fall slightly (T7–FT7 8.5 → 5.9; FC5–FT7 10.3 → 5.7; T7–C5 10.7 → 10.7; FC5–C5 13.5 → 11.5). The left lateral view is therefore near-identical across conditions; the right lateral view differs.

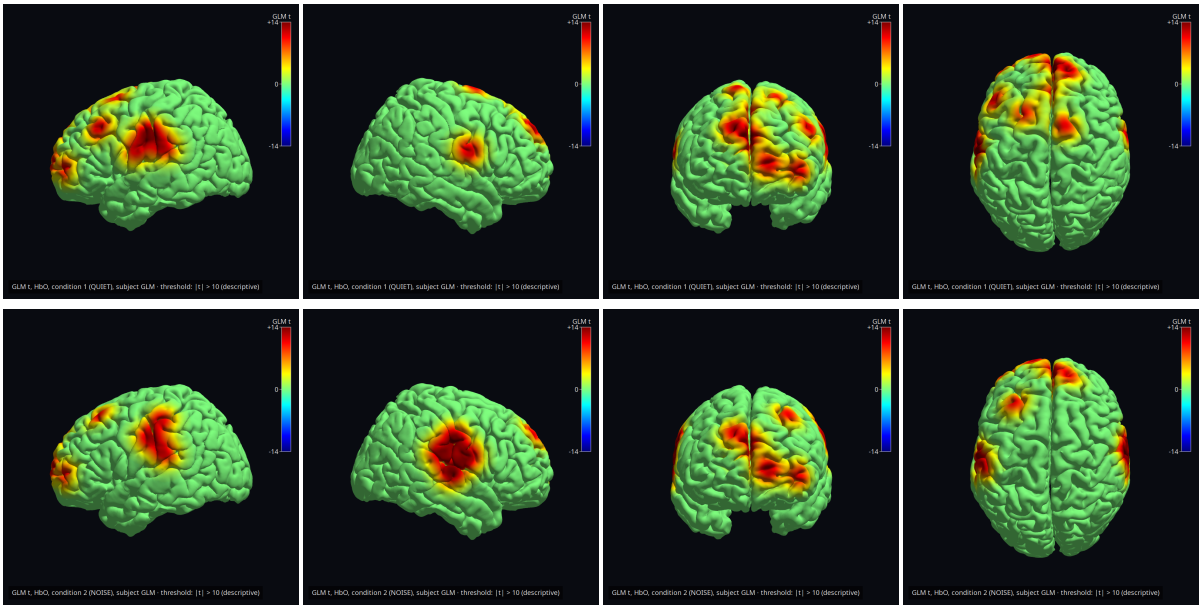


Figure 4: The same maps thresholded at $|t| > 10$ (descriptive). Top: QUIET, bottom: NOISE. The right temporal–premotor cluster (T8–FT8, FC6–C6, T8–C6, FC6–FT8) is present only in NOISE (right lateral and superior views); the left lateral view is essentially unchanged.

Table 1: HbO channels with $t \geq 10$ in either condition (subject GLM, df = 21 392; all $q < 10^{-6}$). β in μM. Bold: channels that cross the $t = 10$ cut in one condition only. Hemi.: left (L), right (R) or midline (M), as assigned by NIRWizard's laterality region of interest (midline-anchored channels AF4–AFz and FC2–FCz count as right, AF3–AFz and FC1–FCz as left).

			QUIET		NOISE	
Channel	Sites	Hemi.	β	t	β	t
S16–D15	T8–FT8	R	0.090	5.01	**0.226**	**12.55**
S12–D12	FC6–C6	R	0.148	5.88	**0.319**	**12.54**
S16–D12	T8–C6	R	0.130	7.21	**0.201**	**11.00**
S12–D15	FC6–FT8	R	0.163	12.06	0.173	12.69
S1–D2	Fpz–Fp2	R	0.350	10.72	0.391	11.94
S4–D3	AF4–AFz	R	0.235	12.19	0.197	10.08
S11–D9	FC2–FCz	R	**0.148**	**10.75**	0.082	5.93
S1–D3	Fpz–AFz	M	0.336	13.30	0.303	11.86
S6–D3	Fz–AFz	M	**0.188**	**10.87**	0.154	8.82
S9–D11	FC5–C5	L	0.289	13.47	0.250	11.55
S1–D1	Fpz–Fp1	L	0.350	11.84	0.379	12.75
S2–D1	AF7–Fp1	L	0.410	11.28	0.395	10.80
S15–D11	T7–C5	L	0.148	10.75	0.149	10.68
S7–D4	F3–F5	L	**0.182**	**12.12**	0.103	6.79
S9–D13	FC5–FT7	L	**0.205**	**10.29**	0.115	5.72
S10–D5	FC1–F1	L	**0.151**	**10.19**	0.110	7.36
S7–D5	F3–F1	L	0.053	6.77	**0.084**	**10.66**
S13–D11	C3–C5	L	0.204	9.85	**0.221**	**10.50**

2.3 Laterality

NIRWizard's Seghier index over the two hemispheres, computed from the HbO t values, gives a laterality index of LI = +0.121 in QUIET ($L = 8.10$, $R = 6.35$) and +0.057 in NOISE ($L = 7.52$, $R = 6.70$): mildly left-dominant while playing in quiet, and close to symmetric in noise. The shift is driven by the right lateral channels above; the left-hemisphere mean falls slightly.

2.4 Data quality

Windowed quality control (QC; SCI ≥ 0.8, PSP ≥ 0.1, 421 windows) flags 27 of 47 channels as *bad* (good-window fraction below 0.75), mostly on the PSP criterion (25 channels; 12 also fail SCI; none fail signal-to-noise ratio, 12–65 dB). The flagged channels are concentrated in the DLPFC/premotor rows and include several of the channels in Table 1: FC6–C6 (0 % good windows), FC6–FT8 (62 %), FC5–C5 (72 %), Fpz–Fp1 (67 %), Fpz–Fp2 (60 %). The temporal channels T8–FT8, T8–C6 and T7–C5 pass; T7–FT7 does not (0 %). These verdicts were *not* applied to the GLM (`qc_applied: false`); a re-run with pruning, or with the PSP threshold relaxed, is needed to see whether the FC6–C6 result survives. Nine short motion segments (2–5 s) were annotated by the PCA step, seven of them in the two NOISE runs.

3 Interpretation and caveats

In this subject, playing in pink noise appears to recruit a right temporal/ventral-premotor region (T8–FT8, T8–C6, FC6–C6, i.e. roughly right superior/middle temporal gyrus and inferior precentral/frontal operculum) that is only weakly engaged in quiet, while the left premotor/motor and prefrontal play response is unchanged. This is consistent with the study's premise that background noise adds an auditory-monitoring load on top of the motor task. Two properties of the setup argue for caution before calling it a noise effect: the loudspeaker stood on the player's *left*, so the added right-temporal signal is on the side contralateral to the noise source and could in part be a direct auditory response to the noise rather than to playing-in-noise; and no QUIET>NOISE / NOISE>QUIET contrast was fitted. Further limitations:$n = 1$ (12 blocks per condition); no short-separation channels or electrocardiography, so systemic contributions cannot be regressed out; HbR shows no corresponding decrease in the right temporal channels (t between -0.2 and $+3.3$), which weakens the case for a purely neuronal origin; and QC was not applied. For further study the next steps are short-separation channels.